AFEKA The Academic College of Engineering in Tel Aviv
Department of Electrical Engineering

Project name:

# Generative image models for enhancing license plate images captured at extreme viewing angles

Students:
Orpaz Ben Aharon
Igor Adamenko

Mentor:
Dr. Sasha Apartsin

# Abstract

Some applications require processing images taken by cameras at extreme viewing angles, resulting in distorted and noisy images.

This study investigates and compares advanced deep-learning models to enhance the clarity and readability of license plate numbers.

It provides insights into the strengths and limitations of each model and contributes to understanding their applicability.

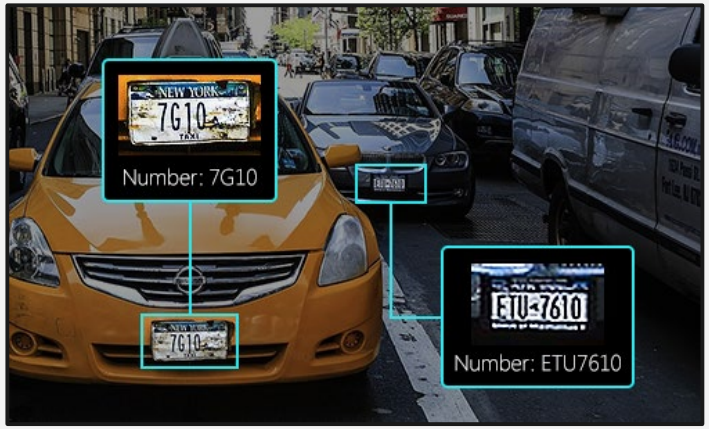
# Introduction

LPR systems are designed to capture license plates under controlled conditions and perform well in those scenarios.

However, challenges arise when using images captured by cameras not intended for LPR, such as ATM cameras used in forensics.

These images are often noisy and distorted, making license plates hard  or impossible to recognize.

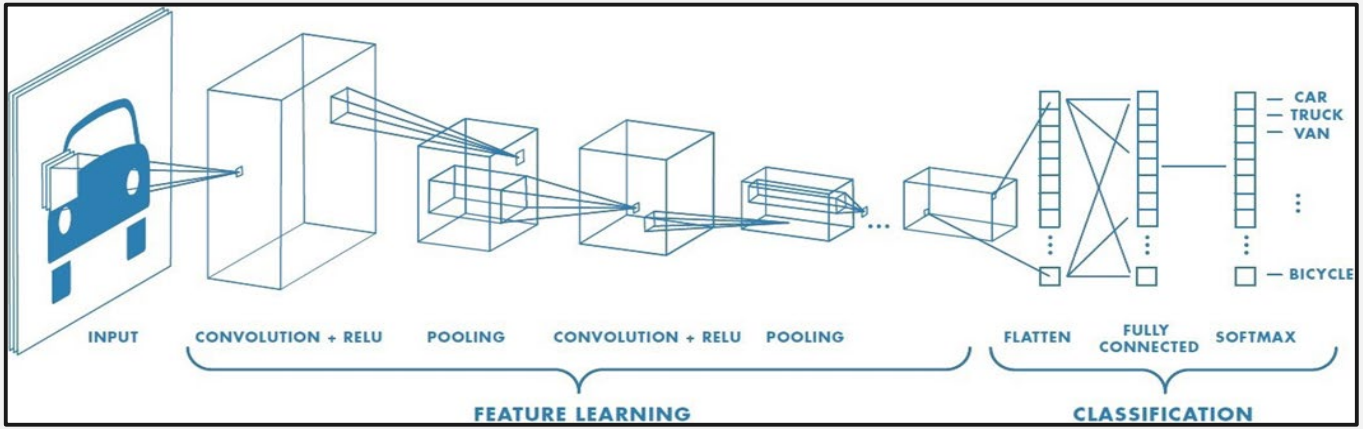
In this study, we aim to find out under what conditions (angles and noise) different deep  -learning models can recover license plates previously unrecognized by humans.



# Introduction

We compare a convolutional neural network UNet with generative AI models, specifically Generative Adversarial Networks (GANs) and Differential Diffusion Models (DDMs).

We generate synthetic license plates and perform augmentations such as projective transform (rotation) and noise injection. These augmented datasets will serve as the learning data for our AI models to evaluate their performance in enhancing license plate images.



# Goals, Objectives

**Main Goal:**

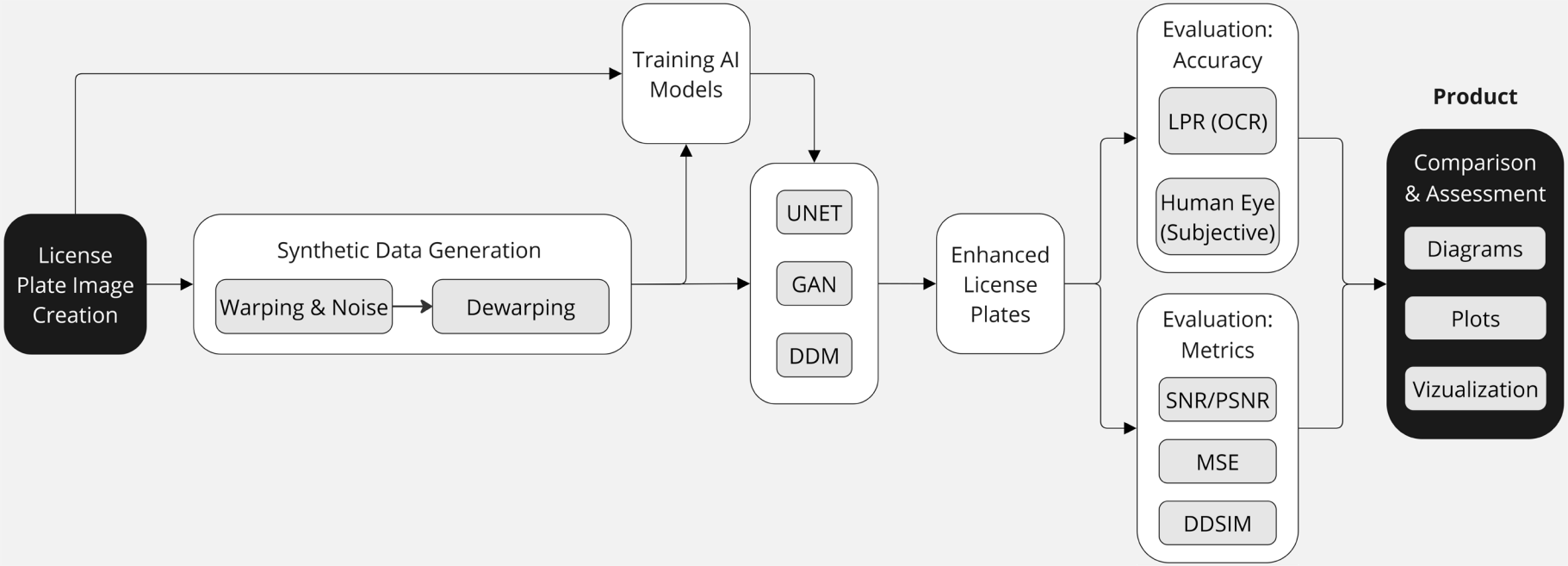
**To compare and assess different deep learning image enhancement methods for improving License Plate Recognition (LPR) accuracy on heavily distorted images.**

**Objectives**

- Create synthetic datasets that simulate various distortions, such as extreme viewing angles and high noise levels.
- Implement and configure different deep learning models, including UNet, Generative Adversarial Networks (GANs), and Differential Diffusion Models (DDMs).
- Conduct training and testing phases to collect and analyze data on recognition accuracy and image quality metrics.
- Conduct a detailed analysis of the strengths and limitations of each model in terms of image enhancement capabilities and computational efficiency.
- Compare the effectiveness of the different models in enhancing heavily distorted images and improving LPR accuracy.

# Metrics

- **SNR Improvement:** Measure improvement in the Signal-to-Noise Ratio (SNR) of the most distorted digit in segmented images.
- **SSIM / DSSIM:** Evaluate image similarity using the Structural Similarity Index (SSIM) and its complement DSSIM.
- **MSE:** Calculate the Mean Squared Error (MSE) between clean and enhanced images to quantify average difference.
- **License Plate Recognition Accuracy:** Compare the percentage of correctly recognized license plate numbers by OpenALPR across clean, distorted, and enhanced images.
- **Confusion Matrix Analysis:** Visualize digit-level true positives, false positives, true negatives, and false negatives.
- **Computational Efficiency:** Measure the time and resources required to process and enhance images for each model.
- **Visual Quality and User Acceptability:** Conduct qualitative evaluations and gather user feedback on the clarity, readability, and perceived accuracy of the enhanced images.

# Synthetic Data Generation and Augmentation

# Project Process Flow